

What is "the Python backend"?

It's the program that powers the **AI Analyst** — the chat feature on your site where a visitor asks a question and gets an answer grounded in your book and platform content. It's called "HTR AI Brain" in the code. That program has to run on some computer, 24/7 or on-demand — **Fly.io is that computer**. That's the entire reason Fly.io exists in this project: it's the rented server running that one Python program.

How does OpenAI relate to it? (Three separate, narrow uses — not "the chatbot")

This is the part worth being precise about, because OpenAI is **not** what writes the chat answers most of the time:

Use	Which AI company	How often	Cost per use
Writing the actual chat replies	Groq (a different company, free tier)	Every chat message	\$0
Finding the right book/platform passage to answer from ("embeddings" — turning your content into a searchable index)	OpenAI (text-embedding-3-small)	Every time content is re-indexed, and apparently on every backend restart (that's the loop you saw in the log)	Fractions of a cent each time
Backup chat writer, only if Groq is completely down	OpenAI (gpt-4o-mini)	Rare/never in normal operation	Small, only when triggered
Text-to-speech for the "personalized learning" audio feature	OpenAI TTS	Only when that specific feature is used	Higher per-use than the above
Chat replies for "advisory/admin" tier users specifically	Anthropic (Claude) — a third company	Only for that one user tier	Separate billing entirely

So: OpenAI's job here is small and specific — turning text into a search index, plus occasional speech audio. It is not the thing answering most users' questions. That's why the app **kept running and answering chat questions fine** even with OpenAI at zero credits — only the "semantic search" indexing step failed, per that log line.

Money — Fly.io

Confirmed, not estimated: the machine spec is 1 shared CPU, 1GB RAM, in Fly's `sjc` (San Jose) region.

- **Before today:** ~\$5.70/month — it was configured to stay on 24/7.
- **After the change I made and you deployed:** Fly bills by actual seconds the machine runs. Your own log proves it's now shutting off after ~8 minutes idle. Realistic range is **roughly \$1-3/month**, and could be less if the chat isn't used often. I can't see your actual Fly invoice — this is the published Fly pricing applied to this exact machine config, not a bill I've read.

Money — OpenAI

I can't tell you a dollar figure to add, because **I can't see your OpenAI account balance or billing history** — only that the log shows it hit zero. What I can tell you:

- The embeddings usage (the main OpenAI cost driver) is extremely cheap per use — a few thousand words of text costs a small fraction of a cent to index.
- The reason it's exactly at **zero right now** is something I genuinely don't know without you checking — it could be a small prepaid amount that's been drawn down over time by repeated re-indexing, or an account that was never funded past a trial. That's at **platform.openai.com/settings/organization/billing**, not something visible from the code.
- TTS (the audio feature) is the one piece here that's meaningfully more expensive per use than the embeddings — if that feature gets real traffic, it's worth watching specifically.

Bottom line: Fly.io is fully explained and now cheap. OpenAI's role is small (search indexing + occasional audio), but I can't tell you what to add without you looking at your own OpenAI billing page — I have no visibility into that account.